Classification of bacteria into gram positive and gram negative based on the cell wall composition.

Gram positive cocci in chains	–	Streptococcus pyogenes
Gram positive cocci in Clusters	–	Staphylococcus aureus
Gram negative Cocci in pairs	–	Neisseria meningitis
Gram positive Bacilli in chains other bacteria	–	Bacillus anthracis
Acid fast bacilli	–	Mycobacterium tuberculosis
Endospore forming	–	Clostridium tetani
Pleomorphic	–	Corynebacterium diphtheria
Gram Negative Bacilli	–	Escherichia coli

When bacteria is present in our body in the absence of disease is called as colonizer. However people can get infection from Pathogenic bacteria through contaminated water, food and air

Skin infection:-

The organism most commonly found in the skin and mucous membrane. It cause superficial and systemic infections. **e.g. Staphylococcus aureus**

Superficial: Boils, impetigo, foliculitis, Pneumonia, Food poisoning, bacteremia.

Respiratory tract infection: - The organism which is more found in the mouth as a normal flora. The infection may be in the upper tract or lower respiratory tract.

Ex. Streptococcus pyogenes

Upper Respiratory tract	–	Sore throat Laryngitis Pharyngitis
Lower Respiratory tract	–	Pneumonia and Tuberculosis



ENZYME LINKED IMMUNO SORBENT ASSAY - ELISA. This technique is used to detect antibodies of HIV Infection.

Gastro intestinal infection: -

Many different species of gram negative bacilli normally found in the intestinal tract. It cause inflammation of the gastrointestinal tract involving both stomach and the small intestine. Symptoms include diarrhoea, vomiting, and abdominal pain.



All the bacteria in our body collectively weigh more than 4 pounds [2kg-bacteria]

Genitourinary tract infection: -

A urinary tract infection (UTI) is an infection in any part of urinary system kidney, uterus, bladder, urethra. However, serious consequences can occur if a UTI spreads to your kidney.

Lower urinary tract	–	Cystitis
Upper urinary tract	–	Pyelonephritis

The most commonly UTI causing organism is **Escherichia coli**

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